MATH 151 Things to Think About for EXAM 2 on Thursday

To prepare for Exam 2, take the following actions:

1. Explain each part of the integrals $\int_C f(x,y)\,ds$, $\int_C \mathbf{F}\cdot d\mathbf{r}$, and $\int_C \mathbf{F}\cdot\hat{\mathbf{n}}\,ds$ in terms of how you would go about evaluating them directly. (What is typically given and what do you have to do?)
2. Demonstrate by evaluating $\displaystyle\int_C \frac{x}{1+y^2}\,ds$, where C is the line segment from (1,0) to (3,1).

   $\sqrt{5}\left(\frac{\pi}{4}+\ln 2\right)$
3. Show that the following is true where C is the triangle in the picture below:

$$\int_C y\,dx - x\,dy = -1$$

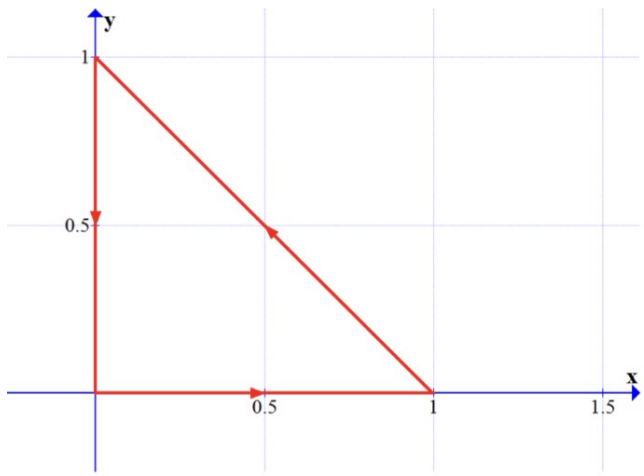

FIGURE 1. The positively oriented curve C

4. If the integral in problem 3 represents the work done by a vector field on a particle moving along the given path C, what is the vector field? If the integral in problem 3 represents the flux of a vector field across the given path C, what is the vector field?
5. Given the conservative vector field $\mathbf{F}(x,y) = 3y\mathbf{i} + 3x\mathbf{j}$, find the work done by $\mathbf{F}$ along the path C given by $\mathbf{r}(t) = \langle t^3 - t + 2, 3 - \sqrt{t-1}\rangle, 1 \le t \le 2$.
6. Discuss the values of the flux integral and work integral for each of the following vector fields (positive, negative, or zero) where C is the path with indicated orientation.

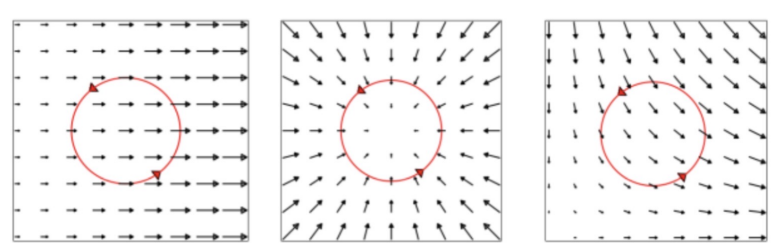

FIGURE 2. Three Vector Fields

7. Homework problems in chapters 9-19 (review the ones in 18 and 19)...